Online Resource 1: Evidence, inference and replication notes Soft-to-Rigid: Shape Continuation as an Alternative Search for Packing Congruent Shapes

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Evidence provenance. This supplement accompanies the controlled-study article and uses the fixed study commit `07e3cbd3c4855e5e19344885fbf672c3d7a0f039`. The source packet contains the input manifest, saved-output verifier and detailed audit. Historical campaign reports and the public arXiv v1 remain separately identified.

1 What the evidence supports

Shape evolution and rounded particle families precede this study. The present contribution is a finite-container, prescribed-target heuristic comparison, with mixed results and explicit failure boundaries. Equal gradient step counts, reference-based baseline pair-evaluation budgets and separate tightening must not be conflated with equal total CPU time. The external-comparison audit found geometry-compatible polygon subsets of Berthold et al., but no matched external performance experiment was conducted. Sparrow's fixed-width strip objective is different. The manuscript claims neither external competitiveness nor causal mechanism, equivalence, a general selector or new controlled-study records.

2 Fixed evidence map

- C00–C10: original campaign plans, raw outputs, selected decimated replays, polishing caches and the serialized event export. The export was inspected as data, without an ActiveGraph runtime or ledger rebuild.
- C11–C16: versioned campaign files and saved analysis outputs, including `analysis/ablations.json`, `C14.json` and `C15.json`. These later campaigns are absent from the original event export, so a universal event-log provenance claim would be inaccurate.
- Manuscript tables and figures retain the native generated macro/table snapshots at the study commit. New audit tables here are generated only by `scripts/verify_joh_packet.py` from saved evidence.
- `submission/joh-20261102/verification/input-manifest.json`: exact input paths, sizes and SHA256 hashes. `report.json`: bounded verification scope and computed checks. `C15-policy-crosswalk.json`: selected policy/run/value identities.

The source repository contains the full run files. This curated supplement provides pointers and selected verification artifacts rather than duplicating the entire campaign corpus.

3 Registered tests, exploratory analyses and reuse

comparison	first lower/only	second lower/only	two-sided p
C10 mixed vs doubled rigid (solved)	4	3	1
C15 pilot vs rigid (size)	2	3	1
C16 new-only harden vs rigid (size)	3	0	0.25

C10 compares 32 rigid plus 32 hardening runs against 64 rigid runs. Its registered one-sided test and the journal's descriptive two-sided test are both disclosed; the observed near-tie is not equivalence. The original decision and report are preserved as historical records; their stronger equivalence-like wording is not supported by the observed near-tie.

C15's primary endpoint compares lowest sizes after the specified policy selection and tightening. A reached-count comparison is a different endpoint. One geometry smoke test per family/path at $n = 8$, seed 3, preceded the written plan; the families were otherwise withheld from tuning and method analysis. The tightened-pilot check is post hoc. Absence of a favorable family prevents a general inference about a selector's ability to find one.

C16 is a targeted squares follow-up. Hardening and rigid runs at $n = 19, 28, 29$ repeat C09 exactly. The bounded verifier checks all 96 such rows by objective, pose, steps and pair evaluations; they are not new independent confirmations. New-only inference excludes these three instances.

The benchmark instances are designed, related geometries. Multiple comparisons, post-selection, ties and catalogue tolerances limit general inference. Bootstrap sensitivity from a single seed pool is not repeated-batch reliability. Mechanism measurements on selected best replays remain descriptive, with contact geometry/forces coupled to orientation.

4 Tuning and tightening scope

method	selected parameters	grid boundary
harden	mu=0.16, noiseScale=8	both
grow	mu=0.08, gamma0=0.1	both
rigid	mu=0.16, noiseScale=8	both
sa	saT0=0.001, mu=0.02	both
pc	pcKick=0.05, pcRelax=19200	both

C06 used one nine-configuration grid for each method, selecting the lowest-of-16 score over eight held-out instances larger than the main test set. All selected settings are on both axis boundaries. Comparable tuning effort is not proof of comparable tuning adequacy. The study retains this limitation; grids were not expanded after observing results.

C14's equal tightening covers a fixed 28-instance subset and three lowest legalised runs per method, with no eligibility gate. The saved cache is sufficient for checking this subset without a solver call. It does not establish robustness for all 188 instances. C16's separate tightening check concerns four selected hardening-only outcomes and remains post hoc.

Recorded simulator timings exclude tuning, numerical tightening and verification. Historical hardware and dependency versions are not fully specified. No current machine identity is substituted for the historical

study environment. The manuscript therefore makes only the scope-limited timing statement supported by the stored diagnostics.

5 Numerical witnesses versus exact certification

There are no below-catalogue claim folders in the controlled-study tree. Numeric reached counts mean the saved size passes the analysis's catalogue tolerance; they do not prove exact equality, optimality or record status. The current exact 2D verifier covers circles, triangles, squares and hexagons, not pentagons/octagons.

n	stored contact objective	checked enlarged side
11	3.8770835900	3.8771233609
19	4.8856180832	4.8856679393
17	4.6755300936	4.6755778489
26	5.6213203436	5.6213775568

The four rows identify selected hardening outcomes: $n = 11, 19$ from C07 and $n = 17, 26$ from C16. Each exact certificate concerns an explicitly enlarged witness: centers scaled by 1.000001 and side set to $1.00001L + 0.000001$. Both native floating-point and rational checkers pass on the supplied derived files. These are feasible upper bounds near the contact outcomes, *not certificates at the original saved objective*. No optimization, retuning, new run or new record claim is involved. Input poses, run identifiers, enlargement rule and checker outputs are in the packet. Certification of every contact outcome remains outside the verified scope.

6 Safe replication boundary

The curated route reads fixed JSON/JSONL and executes only `verify.py` and `packing.certify` on four supplied enlarged witnesses. Missing caches cause an assertion failure, never numerical recomputation. It does not import `packing.ledger`, ActiveGraph, a solver or a provider. In contrast, the native decision/refine/post-hoc modules can call tightening on cache miss; a ledger rebuild can trigger behaviors. Those commands must not be substituted for a read-only audit.

The package was prepared using Python's standard library and existing Tectonic tooling. Compilation uses an existing font environment, documented separately. Full original-campaign reproduction requires a specified run design and a historical environment specification; it is not implied by saved-output checks.

7 Availability, rights and disclosures

Code and study artifacts are linked at https://github.com/yoheinakajima/soft-to-rigid/tree/07e3cbd3c4855e5e19344885fbf672c3d7a0f039. Root code license is MIT; public manuscript v1 is CC BY 4.0. Third-party catalogue/witness rights and companion-note permissions require separate treatment. The public bibliography calls the earlier twelve-cube note submitted to Geombinatorics; current editorial status and reuse rights are unconfirmed. The public build log also flags disputed priority attribution for the earlier cube bound, which is unnecessary to this controlled-study contribution. Funding, author relationships and AI assistance are detailed in the manuscript. Claude assisted the original study; Codex assisted journal writing, audits and software checks. Historical model labels and review payloads are not fully verified. AI-assisted software checks and preparation review are not independent human journal review. The source packet documents tool provenance and unknown versions.